

TEST BANK

CHAPTER 4

DEEP-SEA BIOLOGY

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. thermocline	_____→	A. rapid temperature change
___ 2. dysphotic zone	_____→	B. deep-sea region that receives some light
___ 3. parasitic male	_____→	C. angler fish
___ 4. bioluminescence	_____→	D. photophores
___ 5. epifauna	_____→	E. lives on the sediment surface
___ 6. infauna	_____→	F. lives within the sediment
___ 7. semelparous	_____→	G. reproducing only once
___ 8. homeoviscous adaptation	_____→	H. fluid lipids in membranes
___ 9. copepods	_____→	I. chitinous molts
___ 10. Galápagos Rift Vent	_____→	J. elevated H ₂ S in the water

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

1. The base of the food chain for the Galápagos Rift and other marine hydrothermal vents is which one of the following?

a. Bacteria *

b. Hot water

c. Algae

d. Crabs

e. Copepods
2. Most deepwater abyssal animals are members of which one of the following groups?

a. Scavengers *

b. Herbivores

c. Parasites

d. Primary producers

124 Test Bank for Marine Biology

3. The transition zone between the well-lit upper waters and the waters that no sunlight reaches is termed the ____ zone.
 - a. dysphotic *
 - b. aphotic
 - c. photic
 - d. euphotic
 - e. None of the above
4. Organisms that live in the deep waters of the trenches occupy the ____ zone.
 - a. hadal *
 - b. abyssal
 - c. bathyal
 - d. photic
 - e. aphotic
5. Pressure increases approximately ____ atmosphere(s) for each ____ meter(s) in depth in the water column.
 - a. 1, 10 *
 - b. 2, 10
 - c. 1, 100
 - d. 3, 1
 - e. None of the above
6. The region in the water column where temperature changes most rapidly is called the ____.
 - a. thermocline *
 - b. photic zone
 - c. dysphotic zone
 - d. oxygen minimum zone
 - e. None of the above
7. The oxygen minimum zone lies between about ____ and ____ m.
 - a. 500, 1,000 *
 - b. 1,000, 2,000
 - c. 2,000, 3,000
 - d. > 5,000, < 6,000
 - e. None of the above
8. Which of the following materials represent food to deep-sea animals only after being acted upon by bacteria?
 - a. Dead fish *
 - b. Chitinous exoskeletons
 - c. Live fish
 - d. None of the above
9. Deep-sea communities near islands or continents have ____ food than deep-sea communities farther from land.
 - a. more *
 - b. less
 - c. about the same amount of
10. Which one or more of the following materials are not directly accessible by animals and must be acted upon by bacteria first?
 - a. Chitin *
 - b. Wood *
 - c. Marine snow *
 - d. Whale carcasses
11. Which of the following materials may be directly accessed by animals, but also is used (acted upon) by bacteria first?
 - a. Chitin
 - b. Wood
 - c. Marine snow
 - d. Whale carcasses *
 - e. More than one of the above
12. Many mesopelagic fish are ____.
 - a. a deep black color *
 - b. countershaded
 - c. purple or bright red
 - d. blue
13. Bioluminescence in mesopelagic animals can function as a ____.
 - a. lure for prey capture
 - b. species recognition signal
 - c. method of startling the predator
 - d. More than one of the above *
 - e. None of the above
14. Organisms that reproduce only once in their lifetime are called ____.
 - a. semelparous *
 - b. dioecious
 - c. monoecious
 - d. parthenogenetic
 - e. None of the above

15. Which of the following is the principal inorganic energy source for bacteria of hydrothermal vents?
 - a. hydrogen sulfide *
 - b. water
 - c. carbon dioxide
 - d. sulfur
16. Which of the following are worms that completely lack a digestive system and obtain their energy from symbiotic chemosynthetic bacteria that live in a special organ?
 - a. Vestimentiferans *
 - b. Krill
 - c. Sea cucumbers
 - d. Sea lilies
17. Vents and cold seeps depend on primary production generated by _____.
 - a. chemosynthetic bacteria *
 - b. heterotrophic bacteria
 - c. photosynthetic bacteria
 - d. photosynthetic algae
 - e. chloroplast symbionts
18. The life span of individual hydrothermal vents is on the order of _____.
 - a. decades *
 - b. centuries
 - c. millennia
 - d. weeks
 - e. eons
19. Which of the following organisms are not part of the deep scattering layer?
 - a. Whales *
 - b. Lantern fish
 - c. Euphausiid shrimp
 - d. Squids
 - e. Siphonophores
20. The presence of chemolithoautotrophic bacteria and vent animals _____ suggests that this habitat may act as a stepping stone between sulfide-based vent communities.
 - a. at whale falls *
 - b. in the DSL
 - c. in the thermocline
 - d. in the mesopelagic
21. Which of the following is not an adaptation observed in mesopelagic animals?
 - a. heavy hard parts*
 - b. red color
 - c. black color
 - d. tubular eyes
 - e. large mouth
22. The deep-sea pelagic faunal zone excludes which one of the following regions?
 - a. Intertidal *
 - b. Mesopelagic
 - c. Bathypelagic
 - d. Abyssalpelagic
 - e. Hadalpelagic
23. The deep-sea benthic faunal zone excludes which one of the following regions?
 - a. Intertidal *
 - b. Bathyal
 - c. Abyssal
 - d. Hadal
24. Thermal vents appear to have a life span on the order of _____.
 - a. tens of years *
 - b. hundreds of years
 - c. less than one year
 - d. thousands of years

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [True] The transition zone between the well-lit upper waters and the waters that no sunlight reaches is termed the dysphotic zone.
2. [False] The transition zone between the well-lit upper waters and the waters that no sunlight reaches is termed the aphotic zone.
3. [False] The transition zone between the well-lit upper waters and the waters that no sunlight reaches is termed the photic zone.

126 Test Bank for Marine Biology

4. [True] Phytoplankton growth is limited to the well-lit upper waters called the photic zone.
5. [False] Phytoplankton growth is limited to the well-lit upper waters called the dysphotic zone.
6. [False] Phytoplankton growth is limited to the well-lit upper waters called the aphotic zone.
7. [True] All regions of the deep sea have so little light that photosynthesis is not possible.
8. [True] In the deep sea, the number of animals forming CaCO_3 shells decreases with depth.
9. [False] In the deep sea, the number of animals forming CaCO_3 shells increases with depth.
10. [True] Below several hundred meters in depth, the salinity of the deep sea remains fairly constant.
11. [False] Below several hundred meters in depth, the salinity of the deep sea decreases significantly.
12. [False] Below several hundred meters in depth, the salinity of the deep sea increases significantly.
13. [True] With only a few exceptions, no regions of the deep sea are anaerobic at the bottom.
14. [False] With only a few exceptions, no regions of the deep sea are aerobic at the bottom.
15. [True] The deep-sea benthos appears to have a patchy distribution of organisms.
16. [True] A plot of diversity of the deep-sea benthos as a function of depth indicates that the middle depths have more diversity.
17. [False] The oxygen minimum zone is always found at the bottom of the sea.
18. [True] The dark, cold region called the deep sea comprises 90% of the volume of the world's oceans.
19. [True] Fish with so-called twilight vision usually possess an increased density of rods.
20. [False] In squids with dimorphic eyes, the large eye faces downward.
21. [True] Methane is used by some chemolithotrophic (chemosynthetic) bacteria as a source that provides the energy necessary to fix carbon.
22. [True] The area hypothesis argues that the relatively high species diversity of the deep-sea benthos is simply the result of diversity being calculated over a very wide area.
23. [True] Hydrogen sulfide is toxic to most aerobic animals.
24. [True] Seeps are regions where hydrogen sulfide is released from the benthos and then used by chemolithoautotrophic bacteria as an energy source.
25. [False] Vents and seeps have no greater biodiversity than the rest of the deep sea.
26. [True] Vents have a life span measured in tens of years.
27. [False] Seeps have a life span measured in tens of years.
28. [False] The midwater community has been extremely well studied by scientists.
29. [True] Detritus feeders dominate the community structure of the deep-sea benthos.
30. [False] Enzymes are unaffected by the pressure of the deep sea.
31. [True] The deep-sea benthos shows a paucity of suspension feeders.
32. [False] Seamounts are submerged volcanic mountains that rise only about 500 m above the sea floor.
33. [False] Food is abundant in the deep sea.

Short Answer (Fill in the Blank)

1. The dark, cold region called the deep sea composes 90% of the volume of the world's oceans.
2. The upper region of the deep sea where a minute amount of light penetrates is called the dysphotic zone, and below that lies the aphotic zone, which is permanently dark.
3. Organisms that live on the broad plains of the ocean floor reside in the abyssal zone.
4. Usually, the presence of large eyes in mesopelagic fish is correlated with the presence of light organs.
5. Fish with so-called twilight vision usually possess the vision pigment rhodopsin and an increased density of rods.

6. Squids of the family Histoteuthidae possess dimorphic eyes, where one eye is much larger than the other.
7. In squids with dimorphic eyes, the large eye faces upward.
8. To prevent silhouetting, *Abraliopsis* (a particular species of squid) turns on its photophores.
9. Light-producing organs of deep-sea fish are called photophores.
10. Bioluminescence requires the substrate luciferin and the enzyme luciferase to oxidize the substrate.
11. According to the burglar-alarm hypothesis, a blinding flash of bioluminescent light produced by the prey momentarily startles the predator, thereby permitting the prey to escape.
12. The epifauna are those bottom-dwelling organisms which live on or are associated with the surface of the sea's bottom or substrate.
13. The infauna are those bottom-dwelling organisms which live within the bottom substrate.
14. The Stability-time hypothesis argues that the relatively high species diversity of the deep-sea benthos is the result of a highly stable environment that has persisted over a very long time.
15. The Cropper (or Disturbance) theory argues that the relatively high species diversity of the deep-sea benthos is the result of intense predation that permits a large number of species to exist over a wide area.
16. Hydrogen sulfide is used by chemolithotrophic (chemosynthetic) bacteria as an energy source that drives carbon fixation.
17. The deep-sea benthic regions include the bathyal, abyssal, and hadal zones.
18. Bioluminescent animals produce light from structures called photophores.
19. The vestimentiferan worm, *Riftia*, contains chemosynthetic bacteria in its tissues; these symbiotic bacteria provide all the energy needed by the animal.
20. The phenomenon of animals growing larger at increasing depths is called abyssal gigantism.

Essay Discussion Questions

Develop complete answers to these questions.

1. In morphological terms, how are some deep-sea fishes like terrestrial snakes? Discuss.
2. Explore the adaptive significance of the skeletal anatomy of deep-sea fish.
3. What is abyssal gigantism? Evolutionarily speaking, why did it develop? Or is it just a function of growth rates at low temperatures? Discuss.
4. Explain abyssal gigantism in terms of the Red-Queen Hypothesis (or Red-Queen Race).
5. Examine the concept of parallel bottom communities.
6. Discuss the problems associated with sampling the deep sea. How has technology advanced our understanding of the deep sea by improving sampling? Discuss.
7. Discuss the effects of pressure on deep-sea organisms.
8. What are the effects of increased pressure and decreased temperature on organisms that secrete shells made of calcium carbonate?
9. Review the solubility of CaCO_3 as a function of temperature and pressure, and explore the consequences of that solubility to animals that make shells.
10. If mesopelagic invertebrates live in a realm of total darkness, why are so many colored? Discuss.
11. Fish that live in the abyssal and hadal zones often possess reduced eyes or lack eyes altogether. Why is this so? Discuss.
12. Explain how dimorphic eyes function.
13. How is male parasitism in the angler fish an adaptive feature to the fish that possess this characteristic?
14. Why is the deep-sea benthos depauperate in suspension-feeding animals? Discuss.
15. Compare and contrast suspension feeding in shallow, inshore waters and in the deep-sea benthos.

128 Test Bank for Marine Biology

16. How does the Stability-time hypothesis explain the relatively high species diversity present in the deep-sea benthos? Discuss.
17. How does the Cropper theory explain the relatively high species diversity present in the deep-sea benthos? Discuss.
18. How does the Disturbance theory explain the relatively high species diversity present in the deep-sea benthos? Discuss.
19. How does the area hypothesis explain the relatively high species diversity present in the deep-sea benthos? Discuss.
20. Summarize the general characteristics of deep-sea animals. As part of your answer, offer two characteristics that they possess related to their reproduction, development, physiology, and ecology.
21. Why is the deep sea patchy in its distribution of organisms? What practical problems does that condition present researchers who study the deep-sea benthos?
22. Examine the concept of resource partitioning in the pelagic fauna of the deep sea. How is resource partitioning accomplished by the pelagic fauna of the deep sea? Discuss.
23. What particular challenges does the deep sea present to those who wish to explore it? Discuss.
24. Is the region below the level of light penetration in the open ocean always permanently dark?
25. Compare and contrast vertical zonation of the deep-sea region with zonation of mountains.
26. Explain why one would expect to find fewer organisms in the abyssal pelagic zone than in the epipelagic zone.
27. Explain why one would expect to find fewer organisms in the abyssal benthos than in the benthos of the continental shelf.
28. How does pressure influence the biology of deep-sea organisms (for example, enzymes, growth rates, cell morphology, and protein synthesis)? Discuss this phenomenon, providing examples to support your essay.
29. Are organisms in the deep-sea benthos food limited or predator limited? Discuss.
30. What sorts of materials reach the abyssal zone? Discuss both the quantity and the quality of these materials.
31. Why are mesopelagic fish not countershaded? What is the adaptive strategy of countershading?
32. Why do some deep-sea fish have such large eyes if no surface light ever reaches the depth that they inhabit? Contrast these fish with the blind freshwater cave fish. Why, if both get no light, is one blind and the other not? Discuss.
33. How does a tubular eye work? What purpose does it serve?
34. What curious behavior does the midwater squid *Abraliopsis* exhibit? Discuss the significance of this behavior in relation to this animal's anatomy.
35. Why is bioluminescence called cold light? How can light be cold? After all, if you grab onto a 100-watt lightbulb that has been on for a while, you find that it's quite hot!
36. What evidence would suggest that the production of light is very important in deep-sea animals? What purpose(s) does light production serve?
37. Why are there no seasonal patterns of production and reproduction in the deep-sea benthos? Compare and contrast reproduction in shallow tropical, shallow polar, and the abyssal benthos. Explore the patterns exhibited by each.
38. What is the energy source for the animals that live in the Galápagos Rift Zone?
39. How do bacteria metabolize H_2S to get energy? Is H_2S a reduced or oxidized form of sulfur? In relative terms, compare H_2S with glucose as an energy source for bacteria.
40. What is the DSL? Draw a well-labeled figure, and explain what the diel vertical pattern of the DSL looks like. How is it detected? How did this factor cause problems to oceangoing vessels when sonar was first introduced?

41. What is an opportunistic predator? Opportunistic herbivore? What is meant by the term “resource partitioning”?
42. How are vent communities like the alpine communities that are found on the tops of tall mountains?
43. How are reduced sulfides toxic to aerobic respiration in most animals, and how do hydrothermal-vent animals (for example, the vestimentiferan worm, *Riftia*, and the large clam, *Calymene*) circumvent this problem?
44. Why might assessing the value of a deep-sea mineral resource be difficult? Discuss.
45. Offer an explanation of how gastropods living in vent communities are capable of surviving, given that they lack planktotrophic larvae.
46. Discuss why symbiosis is of such importance to the life of vent communities. Offer examples to support your argument.
47. How are vent communities an exception to the generalization that photosynthetic plants are the energetic foundation for all other life in the ocean?
48. In terms of energy sources, how is the deep-sea benthos like the carpet in a restaurant? Speculate as to whether this restaurant would be an ethnic restaurant or not. Discuss.
49. A ship loaded with paper sinks to the bottom of the deep sea, spilling its contents, which slowly settle out. The paper contains a toxic compound. Speculate as to where these toxins may end up.
50. Why do bacteria that are adapted to high pressures have more fluid lipids in their membranes than other bacteria do?
51. Why are there so few invertebrates with thick calcium carbonate shells in the deep sea?
52. Why is the species diversity of the benthos highest at about 2,000 m and lower at shallower and deeper levels? Discuss.
53. If there is no replenishment of oxygen in the abyssal zone, why are there so few places in the deep sea in which the oxygen concentration is depleted?
54. Discuss species diversity in the mesopelagic zone, the bathyal zone, and around a thermal vent.
55. Why are the major hydrothermal-vent fields located on the midoceanic ridges, sites where the seafloor is spreading?
56. Can the changes in community structure that took place in the “Rose Garden” between 1979 and 1985 be equated to the changes that take place during old-field succession?
57. Speculate as to why the protein concentration of crustaceans decreases with depth in the deep sea.
58. Challenge Question: Draw an energy pyramid for the deep-sea benthos. Explain your figure in terms of the trophic dynamics of the deep sea.
59. Challenge Question: Draw an energy-flow diagram for the bathypelagic zone. Explain your figure in terms of the trophic dynamics of this region.
60. Challenge Question: Compare and contrast the deep-sea hydrothermal vents with cold seeps. Develop a table to provide evidence to support your answer.
61. Challenge Question: Compare and contrast the deep-sea hydrothermal vents with the community that develops one decomposing whalebone. Develop a table to provide evidence to support your answer.
62. Challenge Question: Speculate as to why the angler fish has evolved a life history in which the male is very small and parasitic on the female.
63. Challenge Question: When testifying before Congress, you are asked the following question: Given the limited funds for scientific research, should we send astronauts to explore Mars or should we explore the deep sea? Defend your answer.

